

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--



3 Complex Analysis 2024
 A/Prof Elena Berdysheva
 Mita Ramabulana
Final Exam
 12 November 2024



Instructions:

- This question paper consists of **8 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 60 points = 100% (63 points available).

Problem	1	2	3	4	5	6	Σ	Mark
Maximum Points	10	8	12	13	10	10	60	100%
Result								

Problem 1

(10 points)

Find the set of all $z \in \mathbb{C}$ where the function $f(z) = (\operatorname{Re} z)^2$ is differentiable.

Solution.

Let $z = x + iy$ with $x = \operatorname{Re} z$, $y = \operatorname{Im} z$, and let $f = u + iv$ with $u = \operatorname{Re} f$, $v = \operatorname{Im} f$. We have

$$f(z) = (\operatorname{Re} z)^2 = x^2,$$

and therefore

$$u(x, y) = x^2, \quad v(x, y) = 0.$$

We have

$$u_x(x, y) = 2x, \quad u_y(x, y) = 0, \quad v_x(x, y) = 0, \quad v_y(x, y) = 0.$$

The Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$ take the form

$$2x = 0, \quad 0 = 0.$$

They can only be fulfilled if $\operatorname{Re} z = x = 0$. Thus, f is not differentiable at z if z is not of the form $z = iy$ with $y \in \mathbb{R}$.

Now let us consider $z = iy$ with $y \in \mathbb{R}$. We have

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = \lim_{h \rightarrow 0} \frac{(\operatorname{Re} h)^2 - 0}{h} = \lim_{h \rightarrow 0} \frac{(\operatorname{Re} h)^2}{h}.$$

Using the inequality $|\operatorname{Re} h| \leq |h|$, we estimate

$$0 \leq \lim_{h \rightarrow 0} \left| \frac{f(z+h) - f(z)}{h} \right| = \lim_{h \rightarrow 0} \left| \frac{(\operatorname{Re} h)^2}{h} \right| \leq \lim_{h \rightarrow 0} \frac{|h|^2}{|h|} = \lim_{h \rightarrow 0} |h| = 0.$$

It follows that

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = 0.$$

Thus, f is differentiable at any z of the form $z = iy$ with $y \in \mathbb{R}$. (with $f'(z) = 0$).

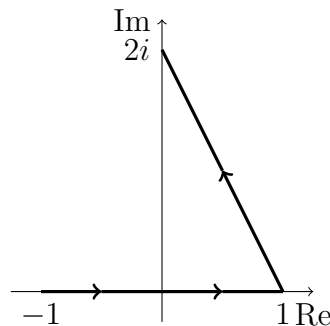
Summarizing, the set of all $z \in \mathbb{C}$ where the function $f(z) = (\operatorname{Re} z)^2$ is differentiable is $\{z \in \mathbb{C} : z = iy \text{ with } y \in \mathbb{R}\}$.

Problem 2

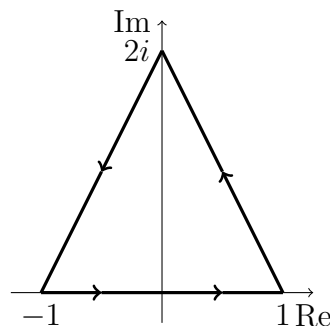
(8 points)

Calculate the following integrals.

- (a) $\int_{\gamma_1} \frac{1}{(z-i)^2} dz$, where γ_1 is the indicated path:



- (b) $\int_{\gamma_2} \frac{1}{(z-i)^2} dz$, where γ_2 is the indicated path:

**Solution.**

- (a) The function $f(z) = \frac{1}{(z-i)^2}$ has an antiderivative $F(z) = -\frac{1}{z-i}$ in the open set $\mathbb{C} \setminus \{i\}$, and γ_1 is a path in $\mathbb{C} \setminus \{i\}$. By the Fundamental Theorem of Calculus,

$$\begin{aligned} \int_{\gamma_1} \frac{1}{(z-i)^2} dz &= F(2i) - F(-1) = -\frac{1}{2i-i} + \frac{1}{-1-i} = -\frac{1}{i} - \frac{1}{1+i} \\ &= i - \frac{1-i}{1^2-i^2} = i - \frac{1-i}{2} = -\frac{1}{2} + \frac{3}{2}i. \end{aligned}$$

- (b) The function $f(z) = \frac{1}{(z-i)^2}$ has one singularity $z_0 = i$ inside γ_2 which is a pole of order 2. By the Residue Theorem

$$\int_{\gamma_2} \frac{1}{(z-i)^2} dz = \text{Res}(f, i) = \frac{1}{1!} \left((z-i)^2 f(z) \right)' \Big|_{z=i} = (1)' \Big|_{z=i} = 0.$$

(Alternatively, one sees immediately that $\text{Res}(f, i) = 0$ because the Laurent coefficient a_{-1} of the function $f(z) = \frac{1}{(z-i)^2}$ is zero.)

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

Problem 3

(12 points)

- (a) Show that there exists no holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ with

$$\left| f' \left(\frac{1}{2} \right) \right| > 4 \max_{|z|=1} |f(z)|.$$

- (b) Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a holomorphic function such that

$$f(z) \neq 0 \quad \text{if} \quad |z| = 1.$$

Put $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. Assume that

$$\int_{\gamma} \frac{1}{f(z)} dz \neq 0.$$

Prove that f has at least one zero in the open unit disc $D(0; 1) = \{z \in \mathbb{C} : |z| < 1\}$.

Solution.

- (a) Consider the unit circle $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. By Cauchy's Integral Theorem for derivatives,

$$f' \left(\frac{1}{2} \right) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{\left(z - \frac{1}{2} \right)^2} dz.$$

We are going to use the Estimation Theorem. If z is on the unit circle, then by the inverse triangle inequality $\left| z - \frac{1}{2} \right| \geq |z| - \left| \frac{1}{2} \right| = 1 - \frac{1}{2} = \frac{1}{2}$. The length of the unit circle is 2π . We obtain

$$\left| f' \left(\frac{1}{2} \right) \right| = \left| \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{\left(z - \frac{1}{2} \right)^2} dz \right| \leq \frac{1}{2\pi} \max_{|z|=1} |f(z)| \cdot \frac{1}{\left(\frac{1}{2} \right)^2} \cdot 2\pi = 4 \max_{|z|=1} |f(z)|.$$

Therefore there is no holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ with

$$\left| f' \left(\frac{1}{2} \right) \right| > 4 \max_{|z|=1} |f(z)|.$$

- (b) By contradiction, suppose that f does not have zeros in $D(0; 1)$. By assumption, f also has no zeros on the boundary of $D(0; 1)$. Since f is not identically zero, by the Identity Theorem its zeros cannot have a limit point. Therefore, the disc $D(0; 2)$ can contain only finitely many zeros of f . It follows that there exists $R > 1$ such that f does not have zeros in $D(0; R)$.

It follows that the function $\frac{1}{f(z)}$ is holomorphic in $D(0; R)$. Then by Cauchy's Integral Theorem

$$\int_{\gamma} \frac{1}{f(z)} dz = 0,$$

which is a contradiction. Therefore, f must have at least one zero in $D(0; 1)$.

Problem 4

(13 points)

- (a) Determine the singularities of the function $f(z) = \frac{1}{z^4 - 1}$ in $\mathbb{C}$ and their type. For poles, give the order. Justify your answer.
- (b) Calculate the residue at each of the poles of f .
- (c) Calculate P.V. $\int_{-\infty}^{\infty} \frac{1}{x^4 - 1} dx$.

Solution.

- (a) The zeros of the polynomial $z^4 - 1$ are 4th roots of 1, namely,

$$z_0 = 1, \quad z_1 = i, \quad z_2 = -1, \quad z_3 = -i.$$

They are all simple poles. Indeed,

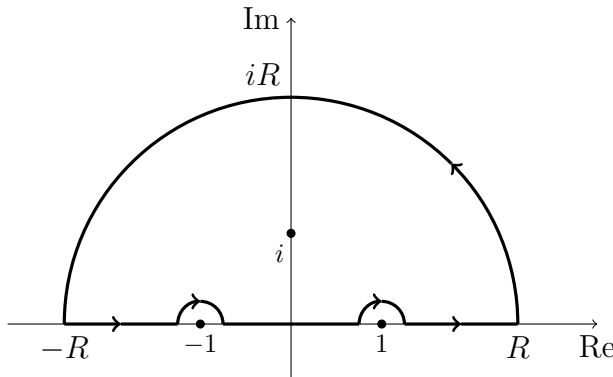
$$f(z) = \frac{1}{(z - z_0)(z - z_1)(z - z_2)(z - z_3)}.$$

For each $j \in \{0, 1, 2, 3\}$, we have $f(z) = \frac{\varphi_j(z)}{z - z_j}$, where φ_j is holomorphic in some open disc centered in z_j , and $\varphi_j(z_j) \neq 0$.

- (b) The residues are

$$\begin{aligned} \operatorname{Res}\left(\frac{1}{z^4 - 1}, 1\right) &= \frac{1}{4z^3}\Big|_{z=1} = \frac{1}{4}, & \operatorname{Res}\left(\frac{1}{z^4 - 1}, i\right) &= \frac{1}{4z^3}\Big|_{z=i} = \frac{i}{4}, \\ \operatorname{Res}\left(\frac{1}{z^4 - 1}, -1\right) &= \frac{1}{4z^3}\Big|_{z=-1} = -\frac{1}{4}, & \operatorname{Res}\left(\frac{1}{z^4 - 1}, -i\right) &= \frac{1}{4z^3}\Big|_{z=-i} = -\frac{i}{4}. \end{aligned}$$

- (c) Consider the path γ indicated below with R large enough:



Let $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$ be the large semicircle with radius R , and let $\gamma_1(t) = 1 + \varepsilon e^{i(\pi-t)}$, and $\gamma_{-1}(t) = -1 + \varepsilon e^{i(\pi-t)}$, $t \in [0, \pi]$, be small semicircles with radius $\varepsilon > 0$ and centers 1 and -1 respectively, negatively oriented.

We have

$$\int_{\gamma} \frac{1}{z^4 - 1} dz = 2\pi i \operatorname{Res}\left(\frac{1}{z^4 - 1}, i\right) = 2\pi i \frac{i}{4} = -\frac{\pi}{2}.$$

Student Number:									
------------------------	--	--	--	--	--	--	--	--	--

On the other hand,

$$\int_{\gamma} \frac{1}{z^4 - 1} dz = \left(\int_{\gamma_R} + \int_{[-R, -1-\varepsilon]} + \int_{\gamma_{-1}} + \int_{[-1+\varepsilon, 1-\varepsilon]} + \int_{\gamma_1} + \int_{[1+\varepsilon, R]} \right) \frac{1}{z^4 - 1} dz.$$

If $|z| = R$, then $|z^4 - 1| \geq |z|^4 - 1 = R^4 - 1$. We estimate

$$\left| \int_{\gamma_R} \frac{1}{z^4 - 1} dz \right| \leq \frac{1}{R^4 - 1} \pi R \rightarrow 0, \quad R \rightarrow \infty.$$

Furthermore,

$$\lim_{\varepsilon \rightarrow +0} \int_{\gamma_1} \frac{1}{z^4 - 1} dz = -\pi i \operatorname{Res} \left(\frac{1}{z^4 - 1}, 1 \right) = -\frac{\pi i}{4},$$

$$\lim_{\varepsilon \rightarrow +0} \int_{\gamma_{-1}} \frac{1}{z^4 - 1} dz = -\pi i \operatorname{Res} \left(\frac{1}{z^4 - 1}, -1 \right) = \frac{\pi i}{4}.$$

It follows that

$$\begin{aligned} \text{P.V.} \int_{-\infty}^{\infty} \frac{1}{x^4 - 1} dx &= \lim_{\substack{R \rightarrow \infty \\ \varepsilon \rightarrow +0}} \left(\int_{[-R, -1-\varepsilon]} + \int_{[-1+\varepsilon, 1-\varepsilon]} + \int_{[1+\varepsilon, R]} \right) \frac{1}{z^4 - 1} dz \\ &= -\frac{\pi}{2} - \left(-\frac{\pi i}{4} \right) - \frac{\pi i}{4} = -\frac{\pi}{2}. \end{aligned}$$

Student Number:									
------------------------	--	--	--	--	--	--	--	--	--

Problem 5

(10 points)

- (a) Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{n^n}{n!} z^{2n-1}$.
- (b) Determine the Laurent series of the function $f(z) = \frac{e^{2z}}{(z+1)^2}$ about the point $z_0 = -1$. Give an explicit formula for the coefficients a_n , $n \in \mathbb{Z}$, of the Laurent series. Give the region of convergence of this series.
- (c) Use the Laurent series in (b) to determine the residue of the function $f(z) = \frac{e^{2z}}{(z+1)^2}$ at $z_0 = -1$.

Solution.

- (a) We are going to use the Ratio Test for the real series $\sum_{n=1}^{\infty} \frac{n^n}{n!} |z|^{2n-1}$. We have

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1} |z|^{2n+1}}{(n+1)!} \cdot \frac{n!}{n^n |z|^{2n-1}} = |z|^2 \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{(n+1)n^n} \\ &= |z|^2 \lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n} = |z|^2 \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = |z|^2 e < 1 \iff |z| < \frac{1}{\sqrt{e}}. \end{aligned}$$

The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{n^n}{n!} z^{2n-1}$ is $R = \frac{1}{\sqrt{e}}$.

- (b) We have

$$\begin{aligned} f(z) &= \frac{e^{2z}}{(z+1)^2} = \frac{1}{(z+1)^2} e^{2(z+1)} e^{-2} = e^{-2} \frac{1}{(z+1)^2} \sum_{n=0}^{\infty} \frac{2^n (z+1)^n}{n!} \\ &= e^{-2} \sum_{n=0}^{\infty} \frac{2^n (z+1)^{n-2}}{n!} = e^{-2} \sum_{n=-2}^{\infty} \frac{2^{n+2} (z+1)^n}{(n+2)!}. \end{aligned}$$

The coefficients of the Laurent series are

$$a_n = \begin{cases} \frac{e^{-2} 2^{n+2}}{(n+2)!}, & n = -2, -1, 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

Since the power series of the exponential function converges for all $z \in \mathbb{C}$, the region of convergence of the above Laurent series is $\mathbb{C} \setminus \{-1\}$.

- (c)

$$\text{Res}(f, -1) = a_{-1} = \frac{e^{-2} 2^1}{1!} = 2e^{-2}.$$

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Problem 6

(10 points)

How many zeros does the polynomial $p(z) = z^3 - 4z^2 - 5$ have in the set $\{z \in \mathbb{C} : |z| > 2\}$?

Solution.

We consider $f(z) = -4z^2$ and $g(z) = z^3 - 5$. On the circle $|z| = 2$ we have

$$|f(z)| = 4|z|^2 = 16, \quad |g(z)| \leq |z|^3 + 5 = 8 + 5 = 13.$$

Thus,

$$|g(z)| < |f(z)|, \quad |z| = 2.$$

By Rouché's Theorem, f and $p = f + g$ have the same number of zeros in $\{z \in \mathbb{C} : |z| < 2\}$. Clearly, $f(z) = -4z^2$ has 2 zeros (more precisely, a zero at $z = 0$ of order 2) in $\{z \in \mathbb{C} : |z| < 2\}$. We conclude that p has 2 zeros in $\{z \in \mathbb{C} : |z| < 2\}$.

Since $|g(z)| < |f(z)|$ on the circle $|z| = 2$, we see that $p = f + g$ cannot have zeros with $|z| = 2$.

Since p is a polynomial of degree 3, it has 3 zeros in $\mathbb{C}$. We conclude that p has exactly one zero in $\{z \in \mathbb{C} : |z| > 2\}$.